

A Low-Power Rectenna with 1.5 V DC Output for Wirelessly Powering Sensors

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Abstract—To better apply wireless power transfer technology in the rapidly emerging Internet of Things (IoT), increasing the rectified output voltage of wireless energy harvesting systems is essential for directly powering low-power electronic devices. This design focuses on a low-power rectenna tailored for the 5G frequency band, capable of directly powering sensor devices. The design is based on a gap-coupled patch antenna, efficiently receiving RF energy in the 5G frequency range. The rectifier evolved from a Greinacher rectifier and Cockcroft–Walton charge pumps, which can output a higher DC voltage by the microstrip line adjusting the signal voltage waveforms. It is experimentally verified that the designed low-power rectenna can provide 1.5 V DC voltage at -10 dBm RF input. Electromagnetic waves are collected through the receiving antenna, and the sensor device is directly powered by 3.5 GHz RF low-power for normal operation. The design can completely demonstrate wireless energy transfer applied in IoT technology to expand the application scenarios.

Keywords—high-voltage, rectifier, rectenna, sensors, wireless power transfer

I. INTRODUCTION

Microwave Wireless Power Transmission (WPT) is a technology that transmits electrical energy from one place to another via microwaves. Unlike traditional wired power transmission, microwave energy transmission does not rely on cables or physical connections but rather transmits electrical energy from a transmitter to a receiver in the form of emitted microwave radiation. This technology has a wide range of potential applications, including solar power generation in space, wireless charging of drones, smart city infrastructure, etc. One of the most compelling applications of microwave wireless energy transfer is solar power stations in space. This concept envisages using solar satellites located in Earth's orbit to collect solar energy and then transmit the energy back to Earth via microwaves. Since the Earth's atmosphere absorbs some of the sun's radiation, a space solar plant would avoid this problem by collecting energy around the clock. It could theoretically provide more energy than terrestrial solar power.

With the rise of IoT, there is an explosion of integrated circuits designed for smart homes, healthcare, and environmental monitoring, among others. Distributed wireless power transfer networks can supply power to various sensors, monitoring devices, and IoT devices, effectively reducing battery replacement frequency and enhancing overall system efficiency. At the same time, RF signals from the surrounding environment can also be a source of energy supply, and environment-oriented energy harvesting has become a popular research topic [1][2]. Radio frequency (RF) signals for mobile communication, radios, and WiFi networks are ubiquitous. Rectennas can harvest these RF energies [3][4], and the feasibility of this technique has been demonstrated in many experiments. There are many papers on energy harvesting.

[5][6][7]. Optimizing antennas to harvest more power [5] or designing rectifiers to improve power conversion efficiency (PCE) [6] are the main ways to improve efficiency.

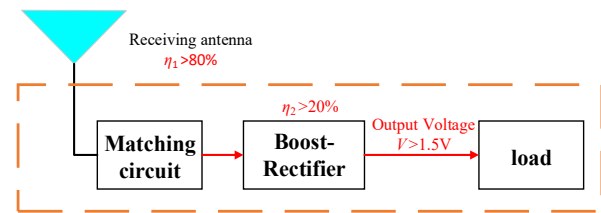


Fig. 1. Receiving System Schematic Diagram.

However, the output voltage characteristics of the rectifiers are rarely considered in most designs. When the RF energy is weak, the output voltage is usually very low, even if the rectifier has a high energy conversion efficiency. Sensors, microelectronic devices, or IoT devices typically operate at a higher voltage than the original voltage received from the wireless energy transfer. They cannot be powered directly by RF energy. Even by back-connecting the boost circuit, the output voltage and DC power of the rectifier must meet the minimum operating voltage and energy consumption needs of the boost circuit. Attempts have also been made in some studies to achieve high-voltage conversion. The half-wave rectifier designed by [8] can output a DC voltage of 0.41 V, which is then boosted to 3.0 V by a boost converter. However, an additional voltage source is required to supply the energy for the boost converter to operate. This lacks practicality. The Greinacher rectifier alone will rectify and boost the output voltage, which is still very low [9]. Dickson charge pumps are also commonly used in voltage boost circuits and must be cascaded to increase the output voltage. The high reverse DC bias and the increase in the number of cascade steps increase the loss of the device, which in turn decreases the efficiency and output voltage[10][11]. An active rectifier consisting of four MOS transistors connected puts a demand on the input voltage when high voltage output is achieved [12][13].

In order to increase the application scenarios of energy transfer systems under low power conditions, increasing the output voltage of the rectifier at low power is the focus of the design. This design focuses on how to enhance the output voltage of the rectifier at low-power RF inputs so that it can power small-power electronic devices for applications such as IoT and other small electronic devices.

II. DESIGN METHODOLOGY

The main objective of this design is to design a low-power wireless energy transfer receiving system operating at the 5G band. Maximum output voltage above 1.5 V is obtained at -10 dBm RF input. It enables direct energy supply to sensor devices for proper operation.

The design consists of two parts, the receiving antenna and the rectifier, as shown in Fig. 1. The rectifier is able to achieve 1.5 V DC output voltage at -10 dBm RF input, which corresponds to the requirement that the receiving efficiency of the designed antenna should be higher than 80%.

A. Design of Receiving Antenna

The output RF energy of the microwave source is transmitted to free space via the transmitting antenna and then received by the receiving antenna, after which it is transmitted to the rectifier at the back end. When the transmit power and the gain of the transmitting antenna are confirmed, the power theoretically received by the receiving antenna can be obtained by the Friis formula [7]:

$$P_r = \frac{G_t G_r P_t \lambda^2}{(4\pi r)^2} \quad (1)$$

where P_r denotes the power received by the receiving antenna, P_t denotes the power emitted by the transmitting antenna, G_r denotes the gain of the receiving antenna, G_t denotes the gain of the transmitting antenna, r denotes the distance between the transmitting antenna and the receiving antenna and λ denotes the antenna's operating wavelength.

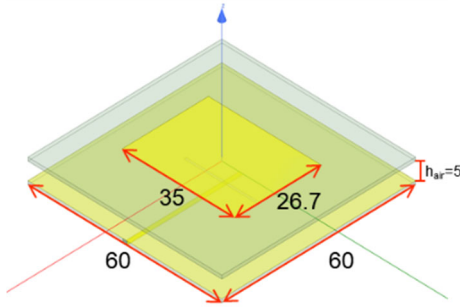


Fig. 2. Antenna shape and size (unit: mm).

The designed receiving antenna should have high receiving efficiency and increase the collected electromagnetic wave energy as much as possible. The operating frequency band of the designed receiving antenna corresponds to the golden frequency band of 5G communication at 3.5 GHz, and the receiving antenna should also have a certain frequency range to receive electromagnetic wave energy near 3.5 GHz.

Microstrip patches are a widely used type of antenna due to their low cost, simplicity of construction, and ease of fabrication. However, the main obstacle to using these antennas in many applications is their limited bandwidth. The slit-coupled feed method is the most common method to expand the bandwidth. Electromagnetic energy is coupled from the feed line to the radiating patch through a gap etched in a metal floor. The advantage of a gap-coupled feed is that the impedance bandwidth of the antenna can be effectively increased, and the shielding effect of the metal floor will make the feed line have very low effects on the radiating patch. Microstrip antennas based on slit coupling are an effective choice for our design due to their compact size, high gain and conversion efficiency and certain operating bandwidths.

A rectangular microstrip antenna is selected as the receiving antenna of the rectifier antenna in this paper, and according to the theory of the rectangular microstrip antenna, the long side

L and the wide side W of the rectangular microstrip antenna can be theoretically and preliminarily calculated by the following empirical equations.

$$L = \frac{c}{2f\sqrt{\epsilon_{re}}} - 2\Delta l \quad (2)$$

$$W = \frac{c}{2f} \left(\frac{\epsilon_r + 1}{2} \right)^{\frac{1}{2}} \quad (3)$$

where ϵ_{re} is the equivalent dielectric constant and Δl is the equivalent radiation gap length. The dielectric substrate used for modeling is F4B-2 polytetrafluoroethylene dielectric sheet with a relative dielectric constant ϵ_r of 2.65, a sheet thickness h of 1 mm, and a loss angle tangent of 0.005. The microstrip patch is fed using gap coupling, and the 3.5 GHz gap-coupled patch antenna shown in Fig. 2 is obtained through the optimization simulation of the software. The antenna's orientation with a maximum gain of 8.7 dB.

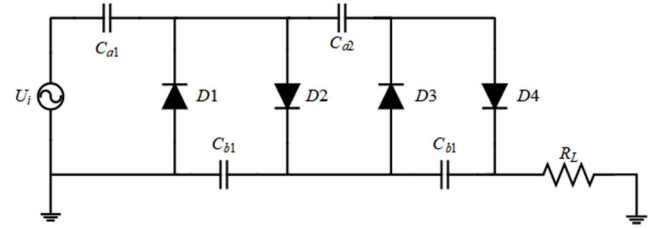


Fig. 3. Basic Cockcroft-Walton circuit structure.

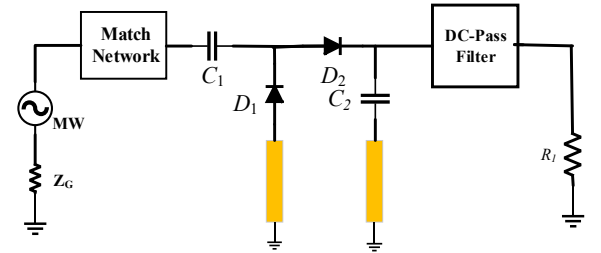


Fig. 4. The schematic diagram of the rectifier of the first order.

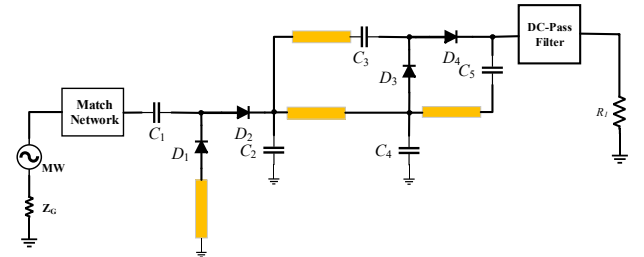


Fig. 5. The schematic diagram of the rectifier of the second order.

B. Boost Rectifier

Cockcroft-Walton charge pumps are widely used in equipment requiring high-voltage DC power supplies due to their simple structure and effective voltage-doubling effect. Through a series of diodes and capacitors composed of a circuit, the use of alternating positive and negative half-cycles of alternating current, and constantly "pump" the charge, to realize the gradual superposition of voltage. The basic Cockcroft-Walton circuit establishes a capacitive step path [14] through all levels of conduction diodes, as shown in Fig. 3.

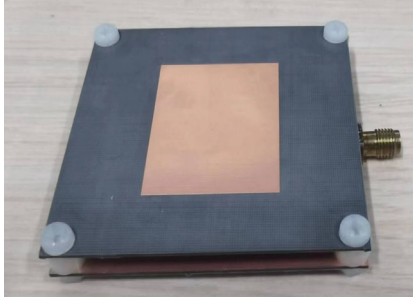


Fig. 6. Antenna photo.

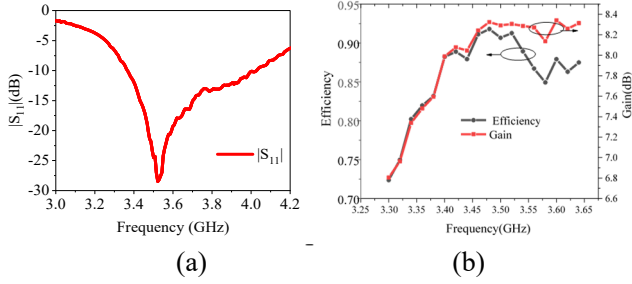


Fig. 7. (a) Antenna bandwidth. (b) Antenna efficiency and gain.

The fundamental principle operates as such: during the negative phase of the RF input voltage, diodes D_1 and D_3 become forward-biased, allowing capacitors C_{a1} and C_{a2} to charge. Conversely, during the positive phase of the RF input voltage, diodes D_2 and D_4 become forward-biased, permitting capacitors C_{b1} and C_{b2} to charge. This process repeats with each cycle, leading to a steady state under no-load conditions where the voltage at the terminals of C_{a1} and C_{a2} stabilizes at U_i , while the voltage at the terminals of C_{b1} and C_{b2} asymptotically approaches $2U_i$.

When a load R_L is introduced in an n -stage basic Cockcroft-Walton voltage multiplier, the voltage-doubling capacitors C_{a1} , C_{a2} , ..., C_{an} are arranged in series to create an RF capacitor stack, and the voltage-doubling capacitors C_{b1} , C_{b2} , ..., C_{bn} are also placed in series to form a DC capacitor stack. During the positive phase of the input voltage, the RF capacitor stack charges the DC capacitor stack. In the subsequent negative phase, the roles reverse, with the DC capacitor stack now charging the RF capacitor stack. Throughout this process, the energy delivered to the load is consistently supplied by the DC capacitor stack.

The rectifier in this design evolved from a Greinacher rectifier and Cockcroft-Walton charge pump to convert RF power to DC while increasing the output voltage. The Greinacher rectifier acts as a rectifier in the first order and forms the initial stage of the Cockcroft-Walton charge pump when connected in multiple orders. Fig. 4 shows the Greinacher rectifier in first order and the schematic diagram of the rectifier in second order is shown in Fig. 5. The microstrip line not only connects the diodes together but also regulates the input voltage to the diodes, the input voltage to the diodes is raised by the microstrip line [15] which helps in increasing the output voltage.

Based on the theory of boost rectifier design, we designed rectifier circuits with orders from 1st order to 4th order. By gradually increasing the voltage, the effect of the order on the rectifier output is explored. Thus, the output voltage target of the rectifier is accomplished.

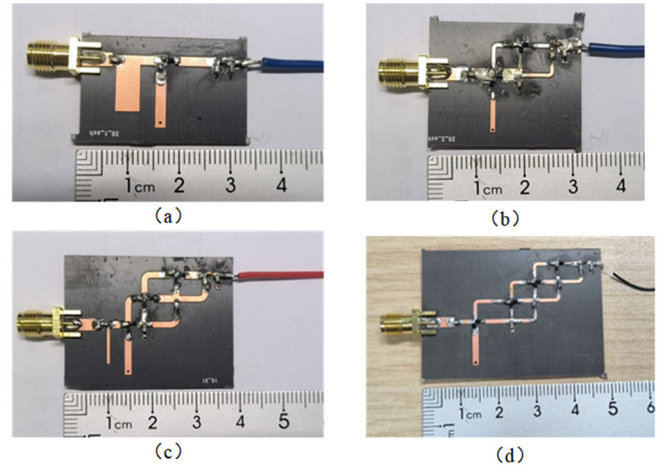


Fig. 8. Rectifiers of different orders (a) first order; (b) second order; (c) third order (d) fourth order.

TABLE I Comparison between different orders rectifiers

Order	Voltage (100k Ω)	Frequency and efficiency	Voltage (330 μ F)
N=1	0.80V	6.4% @3.5GHz	0.91V @3.5GHz
N=2	1.21V	14.5% @3.4GHz	1.22V @3.4GHz
N=3	1.49V	22.4% @3.3GHz	1.55V @3.3GHz
N=4	1.15V	13.1% @3.6GHz	1.59V @3.6GHz

III. RESULTS AND ANALYSIS

A. Antenna measurement

The antenna is machined and designed, and its physical photo is shown in Fig. 6. The characteristics of the designed antenna are measured. Fig. 7(a) shows the S-parameter measurement of the antenna, and it can be seen that the antenna's $|S_{11}|$ is less than -10dB in the frequency range of 3.36GHz to 4.03GHz, which is the antenna's operating bandwidth. The conversion efficiency and gain of the antenna are measured through the antenna chamber, and the results are shown in Fig. 7 (b). It can be seen that the efficiency of the antenna in the frequency band is higher than 80%, up to 91.8%, and in the meantime, the antenna gain is higher than 8 dB.

B. Boost Rectifier measurement

The rectifiers of different orders are processed and designed, and Fig. 8 shows the photographs of the boost rectifiers of orders 1st to 4th. The dielectric substrate used is also F4B-2 PTFE dielectric sheet with a relative dielectric constant ϵ_r of 2.65. SMS7630-0791f is selected for the diode. As the number of orders increases, the output voltage of the rectifier increases and shows different operating characteristics.

The operating characteristics of the four boost rectifiers were measured, and Fig. 10 shows the optimum output voltages and the corresponding conversion efficiencies of the four rectifiers at different input powers. A uniform DC load and capacitive load are considered for voltage measurements in the test, setting the DC load to 100 k Ω , which is similar to the impedance of the selected temperature and humidity sensor. The capacitive load is chosen to be 330 μ F. The maximum output voltage of the first-order rectifier is

measured to be 0.80 V at -10 dBm RF input, and the optimum frequency is 3.5 GHz.

The specific parameter comparisons of the rectifiers of different orders are shown in Table I. From the table, it can be seen that when a 100 k Ω resistor is used as the output load, the rectifier voltage increases gradually with the increase in the number of orders. The voltage reaches a maximum of 1.49 V at the 3rd order, which corresponds to an increase in efficiency to 22.4%. While using a capacitor as a load, due to the higher impedance value of the capacitor, the highest voltage of 1.59 V is measured for the 4th order. The designed 3rd-order boost rectifier can directly supply energy to the thermo-hygrometer, while the 4th-order rectifier can store the energy through the capacitor to supply energy to the device with higher voltage and power consumption. In addition, the measured optimal frequency of the obtained voltages deviates, mainly due to an increase in the number of orders, resulting in a more pronounced error superposition between the devices caused. In the subsequent design, we will optimize the design of the rectifier to reduce the frequency offset, which results in a better matching of the rectifier antenna.

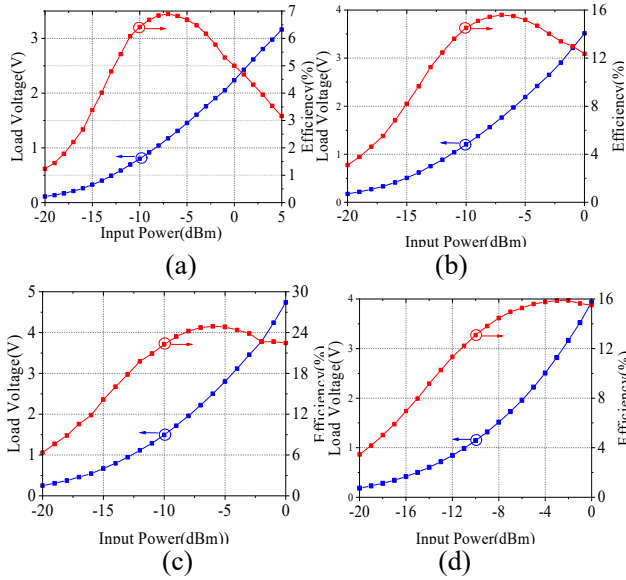


Fig. 9 Rectifier test results (a) first order; (b) second order; (c) third order (d) fourth order.

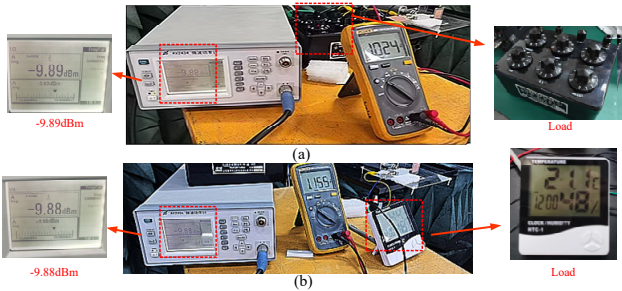


Fig. 10 Measurement. (a) 100 k Ω resistor load (b) Temperature and humidity sensing wirelessly powered 100 μ W RF.

In order to verify the effect of the joint work of the receiving antenna and the boost rectifier, the 3.5 GHz microwave source is used to transmit electromagnetic waves, and the receiving antenna receives the electromagnetic waves at a certain distance and then passes through the output of the boost rectifier, which directly supplies the temperature and humidity

sensors. The electromagnetic wave is transmitted through a horn antenna and received by a slit-coupled antenna at a distance of 1 m. Subsequently, a DC voltage is output through a boost rectifier to supply power wirelessly to the sensors. The experimental test photographs are shown in Fig. 12, where a 3rd order rectifier is used to boost the output. The resistor box is used as a load, and the output voltage at 3.5 GHz is 1.024 V with -10 dBm input power. The sensor is used as a load, and the output voltage is 1.155 V. At this time, the sensor is working properly, and the screen displays the time, temperature, and humidity.

IV. CONCLUSION

In order to better apply wireless power transfer and space energy harvesting technologies to low-power electronic devices, the receiving antenna and the boost rectifier are designed in this design. By designing the boost rectifier with high DC output in cascade with the receiving antenna, a high DC voltage output can be maintained at low power RF input. It can directly power the sensor device for operation. The design discusses the design principles of the receiving antenna and the boost rectifier, and the validity of the design is verified by testing the multistage rectifier. It also successfully ensures the operation of the sensor device by transmitting at -10 dBm RF. This design is a complete demonstration that wireless energy transmission has great application in IoT technology.

ACKNOWLEDGMENT

The National Natural Science Foundation of China (NSFC) programs U22A2015 and 62401382, and the Sichuan Science and Technology Program under Grant 2024YFHZ0282 supported this work.

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